

Speed, Distance and Time — Paper D

HARD • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every answer must carry a unit. An answer with no unit does not score the final mark.
- The mark for each question is shown in square brackets on the right.
- No formula and no conversion factor are given on this paper.
- Where a question asks for an average speed, it means total distance divided by total time.

SECTION 1 – HITTING A TARGET AVERAGE

- 1.** A runner wants to average 5 m/s over a 1000 m race. She covers the first 400 m at 4 m/s.

(a) Calculate the total time she is allowed for the whole race. [1]

(b) Calculate the time she has used, and the time remaining. [2]

(c) Calculate the speed she must run for the rest of the race. [2]

- 2.** Two drivers make different journeys.

Driver A drives for 1 hour at 40 km/h, then for 1 hour at 80 km/h.

Driver B drives 40 km at 40 km/h, then 40 km at 80 km/h.

(a) Calculate Driver A's average speed. [2]

(b) Calculate Driver B's average speed. [2]

(c) Explain why the two answers are different. [1]

SECTION 2 – TWO OBJECTS AT ONCE

3. A car travelling at 25 m/s draws level with a lorry travelling at 20 m/s in the next lane. Calculate how long it takes the car to be 100 m ahead of the lorry. [4]

4. A student shouts towards a cliff and hears the echo 3 seconds later. Sound travels at 340 m/s. Calculate the distance from the student to the cliff. [4]

SECTION 3 – TRAPS AND JUDGEMENT

5. A car travels at 72 km/h for 90 seconds. Calculate the distance it covers, in metres. [4]

6. On a distance-time graph, one object's line is straight and another's curves upwards more and more steeply.

(a) State what each line tells you about that object's motion. [2]

(b) Explain how you can tell which object is moving faster at the very end of the journey. [2]

7. A student has written two answers, and both are wrong. [4]

(i) "The train went 30 km in 20 minutes, so its speed is 1.5 km/h."

(ii) "The flat part of the graph shows the car moving at a constant speed."

For each one, explain the mistake and give the correct answer or description.

End of paper. Check every answer has a unit or a form the question actually asked for.